

## ASSIGNMENT 2 OF 2 · TEAM DECLARATION · DUE FRIDAY 4 SEPTEMBER 2026, 23:59 SGT

# A2 Team Declaration

ONE FILE PER TEAM. Fill this in together and CHECK IT INTO YOUR A2 SUBMISSION FOLDER — do not email it. Keep the file name TEAM\_DECLARATION. A team that files nothing has an incomplete submission. Section 8 of the brief has the full rules.

## 1 · Team and problem

| Team ID | Section | Problem chosen (A or B) |
|---------|---------|-------------------------|
| B-4     | B       | A                       |

Your team is the group you were placed in at the start of the course — A-1 to A-9, B-1 to B-9, C-1 to C-10. Teams are not re-formed for A2.

## 2 · Members

| # | Full name            | Matriculation number |
|---|----------------------|----------------------|
| 1 | SUN YAWEN            | G2604287B            |
| 2 | CHAN HIO WENG        | G2604002G            |
| 3 | LIU ZEYUAN           | G2604153G            |
| 4 | MOHANARANGAN PREETHI | G2604363J            |
| 5 | ZHANG YIZHUO         | G2610294A            |
| 6 | KOU HUILIN           | G2606395C            |

Every member, including anyone who has not yet engaged. Delete unused rows.

## 3 · Repository

**Repository URL:** <https://github.com/ywSun04/PE6201-A2-B-4>

Public, and created by your team. The same URL goes in your final submission.

## 4 · Who owns what

Names against work. The strands below are the suggested split from the FAQ — replace them with your own if you have divided the work differently. Two rows are not optional: everyone writes evaluation cases, and everyone runs one live model. Three models is the floor; N-1 for a team of N is what we expect, with the remaining member taking the v1 pass. Name the model beside each person.

| Strand                                                                                                                                                                                                                                                                                                   | Feeds            | Owner(s)             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------|
| Agent loop and tool implementation. Build the single ReAct loop; implement the tools; gated action = one log record (not a real letter/booking). Extend the loop so several tools can run in one turn; measure sequential vs parallel (turns, tokens) and show correctness does not drop.                | D1, D2(c)        | MOHANARANGAN PREETHI |
| Tool design, descriptors, v1→v2 rewrite. Map how the JSON files join; choose the shortest defensible tool set (three-question table); write a six-field descriptor for every tool; ship a v1 and v2 rewrite and define what changed. Does not write the loop. Freezes the v1 prompt for Member 6 to run. | D2(a), D2(b)     | SUN YAWEN            |
| Guardrails and two reproduced failures. Step cap, budget ceiling, action de-duplication, autonomy gate; ≥ 10 guardrail cases on the scripted backend. Two failures built as deletions from the working agent (one must be a loop failure); restore and show before/after.                                | D3, D7           | CHAN HIO WENG        |
| Evaluation harness and scripted run. Run each case, score against the answer key, log tokens/cost/pass. Submitted code defaults to BACKEND = "scripted" so a marker can clone and reproduce with no API key.                                                                                             | D4, D5(a)        | ZHANG YIZHUO         |
| Fixtures, extra cases, answer key. Coverage plan; add new rows only (EXTRA_*); hand-write expected_outcomes; run check_my_data.py after every change. Never edit or delete the 15 shipped cases.                                                                                                         | D4 (data)        | KOU HUILIN           |
| D0, cost-to-serve, v1 battery, report and demo. Write D0 (why an agent). Three-layer cost model, ledger,                                                                                                                                                                                                 | D0, D6, D5(b) v1 | LIU ZEYUAN           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sensitivity, break-even. Run the v1 prompt on Member 1's exact model (56 trials). Assemble the ≤ 2,000-word report and the 5-minute video (every member speaks). Does not own D7.                                                                                                                                                                                                                                                                                                                                      |       |                                                                                                                                                                                                                                                                                                               |
| Evaluation cases — everyone. 40-case set (15 shipped + ~25 new, roughly 4 new cases each). SUN YAWEN: ASK cases (a named document / test missing).<br>CHAN HIO WENG: escalate by rule + hostile free text.<br>ZHANG YIZHUO: boundary cases (just inside and just outside). KOU HUILIN: duplicates / history.<br>MOHANARANGAN PREETHI and LIU ZEYUAN: ordinary ACT cases and remaining gaps.<br>Every case is labelled from the routing table before the agent runs; new rows only, never editing the 15 shipped cases. | D4    | All six members(KOU HUILIN consolidates into EXTRA_* and the answer key)                                                                                                                                                                                                                                      |
| Live model battery — everyone, one model each. Same eval set and same v2 prompt for all v2 runs. Five families; at least two price tiers.                                                                                                                                                                                                                                                                                                                                                                              | D5(b) | PREETHI — openai/gpt-4o-mini, v2 (cheap)<br>SUN YAWEN — qwen/qwen3.8-flash, v2 (cheap)<br>CHAN HIO WENG — google/gemini-3.8-flash, v2 (mid)<br>ZHANG YIZHUO — claude-haiku-4.5, v2<br>KOU HUILIN — meta/muse-spark-1.3-contributor, v2 (cheap)<br>LIU ZEYUAN — openai/gpt-4o-mini, v1 (same model as PREETHI) |

## 5 · Contribution statement

**THIS SECTION IS NOT OPTIONAL. Do not leave it blank.**

State ONE of the following two things.

| Tick one                            | Statement                                           |
|-------------------------------------|-----------------------------------------------------|
| <input checked="" type="checkbox"/> | All members are contributing.                       |
| <input type="checkbox"/>            | There is a problem. It is described below, by name. |

If you ticked the second box: name the member, what was agreed, and what has happened. Dates help.

Naming a problem here is not an accusation and it does not by itself cost anyone marks. It starts a conversation with a deadline attached: the instructor issues a written warning and the member has nine days to remediate, which usually resolves it. SILENCE AT THIS CHECKPOINT IS READ AS “all members are contributing” — a team that says nothing on 4 September and raises the same problem on 13 September has a much weaker case.

## 6 · Signed off by

Name: Sun Yawen Date: 9.3

One member submits on behalf of the team. Everyone should have read it first.